ATMA RAM SANATAN DHARMA COLLEGE

PRACTICAL FILE

DATABASE MANAGEMENT SYSTEMS

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Course - B.Sc. (Hons) Computer Science

PRACTICALS

1. Retrieve names of students enrolled in any society.
2. Retrieve all society names.
3. Retrieve students' names starting with the letter ‘A’.
4. Retrieve students' details studying in courses ‘computer science’ or ‘chemistry’.
5. Retrieve students’ names whose roll no either starts with ‘X’ or ‘Z’ and ends with ‘9’
6. Find society details with more than N TotalSeats where N is to be input by the user
7. Update society table for the mentor name of a specific society
8. Find society names in which more than five students have enrolled
9. Find the name of the youngest student enrolled in society ‘NSS’
10. Find the name of the most popular society (on the basis of enrolled students)
11. Find the name of two least popular societies (on the basis of enrolled students)
12. Find the students names who are not enrolled in any society.
13. Find the students names enrolled in at least two societies
14. Find society names in which maximum students are enrolled
15. Find names of all students who have enrolled in any society and society names in which at least one student has enrolled
16. Find names of students who are enrolled in any of the three societies ‘Debating’, ‘Dancing’ and ‘Sashakt’.
17. Find society names such that its mentor has a name with ‘Gupta’ in it.
18. Find the society names in which the number of enrolled students is only 10% of its capacity.
19. Display the vacant seats for each society.
20. Increment Total Seats of each society by 10%
21. Add the enrollment fees paid (‘yes’/’No’) field in the enrollment table.
22. Update date of enrollment of society id ‘s1’ to ‘2018-01-15’, ‘s2’ to the current date and ‘s3’ to ‘2018-01-02’.
23. Create a view to keep track of society names with the total number of students enrolled in it.
24. Find student names enrolled in all the societies.
25. Count the number of societies with more than 5 students enrolled in it
26. Add column Mobile number in student table with default value ‘9999999999’
27. Find the total number of students whose age is > 20 years.
28. Find names of students who were born in 2001 and are enrolled in at least one society.
29. Count all societies whose name starts with ‘S’ and ends with ‘t’ and at least 5 students are enrolled in the society.
30. Display the following information:
    Society name Mentor name Total Capacity Total Enrolled Unfilled Seats

II. **Do the following database administration commands:**

Create user, create role, grant privileges to a role, revoke privileges from a role, create index

Creating Database

```sql
CREATE DATABASE college_datab;
USE college_datab;
CREATE TABLE STUDENT (
    RollNo VARCHAR(6) PRIMARY KEY,
    Name VARCHAR(50),
    DOB DATE,
    Course VARCHAR(50)
);

CREATE TABLE SOCIETY (
    SID VARCHAR(6) PRIMARY KEY,
    SName VARCHAR(50),
    MentorName VARCHAR(50),
    TotalSeats INT
);

CREATE TABLE ENROLLMENT (
    RollNo VARCHAR(6),
    SID VARCHAR(6),
    DateOfEnrollment DATE,
    FOREIGN KEY (RollNo) REFERENCES STUDENT(RollNo),
    FOREIGN KEY (SID) REFERENCES SOCIETY(SID)
```

```sql
);
INSERT INTO STUDENT VALUES
('R1','Aman','2001-05-10','computer science'),
('R2','Riya','2000-03-15','chemistry'),
('R3','Karan','2002-07-20','math'),
('R4','Neha','2001-01-25','computer science'),
('R5','Arjun','2000-11-12','physics');

INSERT INTO SOCIETY VALUES
('S1','NSS','Mr Gupta',50),
('S2','Debating','Ms Sharma',40),
('S3','Dancing','Mr Singh',30),
('S4','Sashakt','Mrs Gupta',20),
('S5','Sports','Mr Khan',60);

INSERT INTO ENROLLMENT VALUES
('R1','S1','2024-01-01'),
('R2','S1','2024-01-02'),
('R3','S2','2024-01-03'),
('R1','S3','2024-01-04'),
('R4','S1','2024-01-05'),
('R5','S5','2024-01-06'),
('R2','S2','2024-01-07');
```

1. Retrieve names of students enrolled in any society.

Query:

```sql
SELECT DISTINCT S.Name
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Aman
Riya
Karan
Neha
Arjun

---

2. Retrieve all society names.

Query:

```
SELECT SName FROM SOCIETY;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SName
NSS
Debating
Dancing
Sashakt
Sports

.

---

3. Retrieve students' names starting with the letter 'A'.

Query:

```
SELECT Name FROM STUDENT
WHERE Name LIKE 'A%';
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Aman
Arjun

4. Retrieve students' details studying in courses 'computer science' or 'chemistry'.

```
SELECT * FROM STUDENT
WHERE Course IN ('computer science', 'chemistry');
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import:

RollNo	Name	DOB	Course
R1	Aman	2001-05-10	computer science
R2	Riya	2000-03-15	chemistry
R4	Neha	2001-01-25	computer science
NULL	NULL	NULL	NULL

---

5. Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'

Query:

```
SELECT Name FROM STUDENT
WHERE RollNo NOT LIKE 'X%'
AND RollNo NOT LIKE 'Z%'
AND RollNo LIKE '%1';
```

.

Output:

Result Grid | Filter Rows: | Export:

Name
Aman

---

6. Find society details with more than N Total Seats where N is to be input by the user

Query:

```
SELECT * FROM SOCIETY
WHERE TotalSeats > 30;
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import:

SID	SName	MentorName	TotalSeats
S1	NSS	Mr Gupta	50
S2	Debating	Ms Sharma	40
S5	Sports	Mr Khan	60
NULL	NULL	NULL	NULL

7. Update society table for the mentor name of a specific society Query:

```
UPDATE SOCIETY
SET MentorName = 'Dr Sharma'
WHERE SName = 'NSS';
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

SID	SName	MentorName	TotalSeats
S1	NSS	Dr Sharma	50
S2	Debating	Ms Sharma	40
S3	Dancing	Mr Singh	30
S4	Sashakt	Mrs Gupta	20
S5	Sports	Mr Khan	60

30

8. Find society names in which more than one students have enrolled

Query:

```
SELECT SID
FROM ENROLLMENT
GROUP BY SID
HAVING COUNT(RollNo) > 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SID
S1
S2

9. Find the name of the youngest student enrolled in society 'NSS'

```
SELECT S.Name
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY SO ON E.SID = SO.SID
WHERE SO.SName = 'NSS'
ORDER BY S.DOB DESC
LIMIT 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Name
Aman

---

10. Find the name of the most popular society (on the basis of enrolled students)

Query:

```
SELECT SO.SName
FROM SOCIETY SO
JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName
ORDER BY COUNT(E.RollNo) DESC
LIMIT 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

SName
NSS

---

11. Find the name of two least popular societies (on the basis of enrolled students).

```
SELECT SO.SName
FROM SOCIETY SO
JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName
ORDER BY COUNT(E.RollNo) ASC
LIMIT 2;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

SName
Sports
Dancing

12. Find the students names who are not enrolled in any society.

Query:

```
SELECT Name FROM STUDENT
WHERE RollNo NOT IN (SELECT RollNo FROM ENROLLMENT);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name

---

13. Find the students names enrolled in at least two societies.

```
SELECT RollNo
FROM ENROLLMENT
GROUP BY RollNo
HAVING COUNT(SID) >= 2;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

RollNo
R1
R2

14. Find society name in which maximum students are enrolled.

```
SELECT SO.SName
FROM SOCIETY SO
JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName
ORDER BY COUNT(*) DESC;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SName
NSS
Debating
Dancing
Sports

---

15. Find names of all students who have enrolled in any society and society names in which at least one student has enrolled

```
SELECT S.Name, SO.SName
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY SO ON E.SID = SO.SID;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	SName
Aman	NSS
Aman	Dancing
Riya	NSS
Riya	Debating
Karan	Debating

.

---

16. Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.

```
SELECT DISTINCT S.Name
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
JOIN SOCIETY SO ON E.SID = SO.SID
WHERE SO.SName IN ('Debating', 'Dancing', 'Sashakt');
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Karan
Riya
Aman

---

17. Find society names such that its mentor has a name with 'Gupta' in it.

Query:

```
SELECT SName FROM SOCIETY
WHERE MentorName LIKE '%Gupta%';
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	SName
▶	Sashakt

---

18. Find the society names in which the number of enrolled students is only 10% of its capacity.

Query:

```
SELECT SO.SName
FROM SOCIETY SO
JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName, SO.TotalSeats
HAVING COUNT(E.RollNo) = 0.1 * SO.TotalSeats;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	SName

---

19. Display the vacant seats for each society.

```
SELECT SO.SName,
SO.TotalSeats - COUNT(E.RollNo) AS VacantSeats
FROM SOCIETY SO
LEFT JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName, SO.TotalSeats;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SName	VacantSeats
NSS	47
Debating	38
Dancing	29
Sashakt	20
Sports	59

---

20. Increment Total Seats of each society by 10%

Query:

```
SET SQL_SAFE_UPDATES = 0;
UPDATE SOCIETY
SET TotalSeats = TotalSeats * 1.1;
SELECT * FROM SOCIETY;
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

SID	SName	MentorName	TotalSeats
S1	NSS	Dr Sharma	81
S2	Debating	Ms Sharma	64
S3	Dancing	Mr Singh	48
S4	Sashakt	Mrs Gupta	32
S5	Sports	Mr Khan	97

---

21. Add the enrollment fees paid ('yes'/'No') field in the enrollment table.

Query:

```
USE college_datab;
ALTER TABLE Enrollment
ADD FeesPaid VARCHAR(3) DEFAULT 'No' CHECK (FeePaid IN ('Yes', 'No'));
```

---

22. Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'.

Query:

```
UPDATE ENROLLMENT
SET DateOfEnrollment = CURRENT_DATE
WHERE SID IN ('S1','S2');
SELECT * FROM ENROLLMENT;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

RollNo	SID	DateOfEnrollment	FeesPaid
R1	S1	2026-04-19	NULL
R2	S1	2026-04-19	NULL
R3	S2	2026-04-19	NULL
R1	S3	2024-01-04	NULL
R4	S1	2026-04-19	NULL

---

23. Create a view to keep track of society names with the total number of students enrolled in it.

Query:

```
CREATE VIEW Society_Enrollment AS
SELECT SO.SName, COUNT(E.RollNo) AS TotalStudents
FROM SOCIETY SO
LEFT JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName;
SELECT * FROM Society_Enrollment;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SName	TotalStudents
NSS	3
Debating	2
Dancing	1
Sashakt	0
Sports	1

24. Find student names enrolled in all the societies.

Query:

```
SELECT S.Name
FROM STUDENT S
WHERE NOT EXISTS (
    SELECT SID FROM SOCIETY
    WHERE SID NOT IN (
        SELECT SID FROM ENROLLMENT
        WHERE RollNo = S.RollNo
    )
);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name

25. Count the number of societies with more than 5 students enrolled in it

Query:

```
SELECT COUNT(*)
FROM (
    SELECT SID
    FROM ENROLLMENT
    GROUP BY SID
    HAVING COUNT(RollNo) > 1
) AS temp;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
2

26. Add column Mobile number in student table with default value '9999999999'

Query:

```
ALTER TABLE STUDENT
ADD Mobile VARCHAR(10) DEFAULT '9999999999';
DESCRIBE STUDENT;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
RollNo	varchar(6)	NO	PRI	NULL	
Name	varchar(50)	YES		NULL	
DOB	date	YES		NULL	
Course	varchar(50)	YES		NULL	
Mobile	varchar(10)	YES		9999999999	

---

27. Find the total number of students whose age is > 20 years.

Query:

```
DESCRIBE STUDENT;
SELECT COUNT(*)
FROM STUDENT
WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) >= 20;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
2

.

---

28. Find names of students who were born in 2001 and are enrolled in at least one society.

Query:

```
SELECT DISTINCT S.Name
FROM STUDENT S
JOIN ENROLLMENT E ON S.RollNo = E.RollNo
WHERE YEAR(S.DOB) = 2001;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Aman
Neha

---

29. Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.

Query:

```
SELECT COUNT(*)
FROM SOCIETY SO
JOIN ENROLLMENT E ON SO.SID = E.SID
WHERE SO.SName LIKE 'S%t'
GROUP BY SO.SID
HAVING COUNT(E.RollNo) >= 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)

---

30. Display the following information:
Society name | Mentor name | Total Capacity | Total Enrolled | Unfilled Seats

Query:

```
SELECT SO.SName, SO.MentorName, SO.TotalSeats,
COUNT(E.RollNo) AS TotalEnrolled,
SO.TotalSeats - COUNT(E.RollNo) AS UnfilledSeats
FROM SOCIETY SO
LEFT JOIN ENROLLMENT E ON SO.SID = E.SID
GROUP BY SO.SName, SO.MentorName, SO.TotalSeats;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SName	MentorName	TotalSeats	TotalEnrolled	UnfilledSeats
NSS	Dr Sharma	210	3	207
Debating	Ms Sharma	165	2	163
Dancing	Mr Singh	124	1	123
Sashakt	Mrs Gupta	84	0	84
Sports	Mr Khan	253	1	252